

1 CM

$$\mathbf{f}^* = \mathbf{f} - \underline{\underline{T}}^{-1} \underline{\underline{S}} \underline{\underline{T}} (\mathbf{f} - \mathbf{f}^{\text{eq}}) + \underline{\underline{T}}^{-1} \left(\underline{\underline{I}} - \frac{1}{2} \underline{\underline{S}} \right) \underline{\underline{T}} \Phi, \quad (1.1)$$

where $\underline{\underline{T}}$ is the transformation matrix to central moment space, $\underline{\underline{S}}$ is the diagonal relaxation matrix and Φ is the Guo forcing source term in population space. The rows of the transformation matrix are given as follows,

$$\begin{aligned} T_{0,i} &= 1, & T_{1,i} &= \tilde{c}_{ix}, & T_{2,i} &= \tilde{c}_{iy}, & T_{3,i} &= \tilde{c}_{iz}, \\ T_{4,i} &= \tilde{c}_{ix} \tilde{c}_{iy}, & T_{5,i} &= \tilde{c}_{ix} \tilde{c}_{iz}, & T_{6,i} &= \tilde{c}_{iy} \tilde{c}_{iz}, \\ T_{7,i} &= \tilde{c}_{ix}^2 - \tilde{c}_{iy}^2, & T_{8,i} &= \tilde{c}_{ix}^2 - \tilde{c}_{iz}^2, & T_{9,i} &= \tilde{c}_{ix}^2 + \tilde{c}_{iy}^2 + \tilde{c}_{iz}^2, \\ T_{10,i} &= t_i^{120}, & T_{11,i} &= t_i^{102}, & T_{12,i} &= t_i^{012}, & T_{13,i} &= t_i^{210}, & T_{14,i} &= t_i^{201}, & T_{15,i} &= t_i^{021}, & T_{16,i} &= t_i^{111}, \\ T_{17,i} &= t_i^{220}, & T_{18,i} &= t_i^{202}, & T_{19,i} &= t_i^{022}, & T_{20,i} &= t_i^{211}, & T_{21,i} &= t_i^{121}, & T_{22,i} &= t_i^{112}, \\ T_{23,i} &= t_i^{122}, & T_{24,i} &= t_i^{212}, & T_{25,i} &= t_i^{221}, & T_{26,i} &= t_i^{222}, \end{aligned} \quad (1.2)$$

where $\tilde{c}_{i\alpha} \equiv c_{i\alpha} - u_\alpha$ and $t_i^{abc} = \tilde{c}_{ix}^a \tilde{c}_{iy}^b \tilde{c}_{iz}^c$. The equilibrium is given explicitly in central moment space,

$$t_0^{\text{eq}} = \rho, \quad t_9^{\text{eq}} = 3p, \quad t_{17}^{\text{eq}} = \frac{1}{3}p, \quad t_{18}^{\text{eq}} = \frac{1}{3}p, \quad t_{19}^{\text{eq}} = \frac{1}{3}p, \quad t_{26}^{\text{eq}} = \frac{1}{9}p, \quad (1.3)$$

where $\mathbf{t}^{\text{eq}} \equiv \underline{\underline{T}} \mathbf{f}^{\text{eq}}$. Note that the indices that are not mentioned are zero. The same is done for the Guo forcing source term,

$$\begin{aligned} t_1^\Phi &= F_x, & t_2^\Phi &= F_y, & t_3^\Phi &= F_z, & , \\ t_{10}^\Phi &= \frac{1}{3}F_x, & t_{11}^\Phi &= \frac{1}{3}F_x, & t_{12}^\Phi &= \frac{1}{3}F_y, & t_{13}^\Phi &= \frac{1}{3}F_y, & t_{14}^\Phi &= \frac{1}{3}F_z, & t_{15}^\Phi &= \frac{1}{3}F_z, \\ t_{23}^\Phi &= \frac{1}{9}F_x, & t_{24}^\Phi &= \frac{1}{9}F_y, & t_{25}^\Phi &= \frac{1}{9}F_z, & , \end{aligned} \quad (1.4)$$

where $\mathbf{t}^\Phi \equiv \underline{\underline{T}} \Phi$.